Note

Inclusive series:

In the class-intervals, if the upper limit and lower limit are included in that class interval then it is called inclusive series. For example, 11-20, 21-30 , 31-40, 41-50 etc is an inclusive series.

Here, the data 11 and 20 are included in the class (11-20) and so on. Clearly, it is a discontinuous series.

Exclusive series:

In the class intervals, if the upper limit of one class interval is the lower limit of the next class interval then it is called exclusive series. For example, 10-15, 15-20, 20-25, 25-30 etc., is an exclusive series.

Here, 15 is included in the class 15-20 and 20 is included in 20-30. Clearly, it is a continuous series.

6.2.2 (i) Construction of grouped frequency distribution table – Continuous series

Example 6.2

The EB bill(in ₹) of each of the 26 houses in a village are given below. Construct the frequency table.

215	200	120	350	800	600	350	400	180	210	170	305	204
220	425	540	315	640	700	790	340	586	660	785	290	300

Solution:

Maximum bill amount = ₹ 800

Minimum bill amount = ₹ 120

Range = largest value – smallest value

Range = 800 – 120 = ₹ 680

Suppose if we want to take class size as 100, then

The number of possible class intervals = $\frac{Range}{Class\ size} = \frac{680}{100} = 6.8 \simeq 7$

Class Intervals	Tally Marks	Frequency
100-200	III	3
200-300	𝍸 I	6
300-400	𝍸 I	6
400-500	II	2
500-600	II	2
600-700	III	3
700-800	IIII	4
	Total	26

06_8th_Maths_Statistics_Chap 6.indd 216 20/11/2022 13:21:20